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Machine Learning – Jan 2026

Seminar 5

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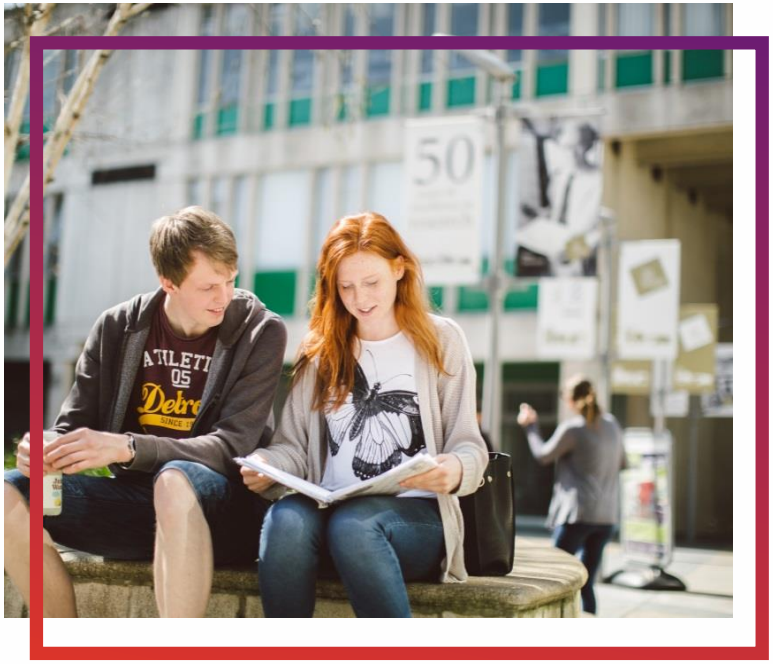
Agenda – Machine Learning

- **Machine Learning**–

Recap module organisation, deadlines, assessments

- Unit 7-8 & Seminar on ANN



Module Structure: Units

Topics to be covered include:

- Unit 1: Introduction to Machine Learning (ML)
- Unit 2: Exploratory Data Analysis
- Unit 3: Correlation and Regression
- Unit 4: Linear Regression with Scikit-Learn
- Unit 5: Clustering
- Unit 6: Clustering with Python
- Unit 7: Introduction to Artificial Neural Networks (ANNs)
- Unit 8: Training an Artificial Neural Network
- Unit 9: Introduction to Convolutional Neural Networks (CNNs)
- Unit 10: CNN Interactive Learning
- Unit 11: Model Selection and Evaluation
- Unit 12: Industry 4.0 and Machine Learning

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Formative and e-Portfolio Activities

We recommend that you participate in all formative activities and make frequent entries to your **e-portfolio which is assessed in this module**.

You will find the main e-portfolio and formative activities below. Please make sure that you read the guidance in each unit to fully understand weekly study requirements.

Also read the full e-Portfolio guidance for the final submission in unit 12.

Unit(s)	Component	Deadline	e-Portfolio/Formative
1 - 3	Collaborative Discussion 1	End of unit 1, 2 and 3	e-Portfolio
2	Unit 2 Seminar Preparation	End of unit 2	Formative and e-Portfolio
3	e-Portfolio Component	End of unit 3	Formative and e-Portfolio
4	Unit 4 Seminar Preparation	End of Unit 4	Formative and e-Portfolio
5	Wiki Activity	End of unit 5	Formative and e-Portfolio
5	e-Portfolio Component	End of unit 5	e-Portfolio
6	Unit 6 Seminar Preparation	End of unit 6	Formative and e-Portfolio
7	e-Portfolio Component	End of unit 10	e-Portfolio
8	Unit 8 Seminar Preparation	End of unit 8	Formative and e-Portfolio
8	e-Portfolio Component	End of unit 8	Formative and e-Portfolio
8 - 10	Collaborative Discussion 2	End of unit 8, 9 and 10	e-Portfolio
9	e-Portfolio Component	End of unit 9	e-Portfolio
10	Unit 10 Seminar Preparation	End of unit 10	Formative and e-Portfolio
11	e-Portfolio Component	End of unit 11	e-Portfolio
12	Unit 12 Seminar Preparation	End of unit 12	Formative and e-Portfolio

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Python Introduction and setup

https://www.kaggle.com/

JupyterLab

https://jupyter.org/try

How to use Jupyter with R

https://www.python.org/

https://anaconda.cloud/

Welcome to Colaboratory

File Edit View Insert Runtime Tools

Table of contents

Getting started

https://colab.research.google.com/

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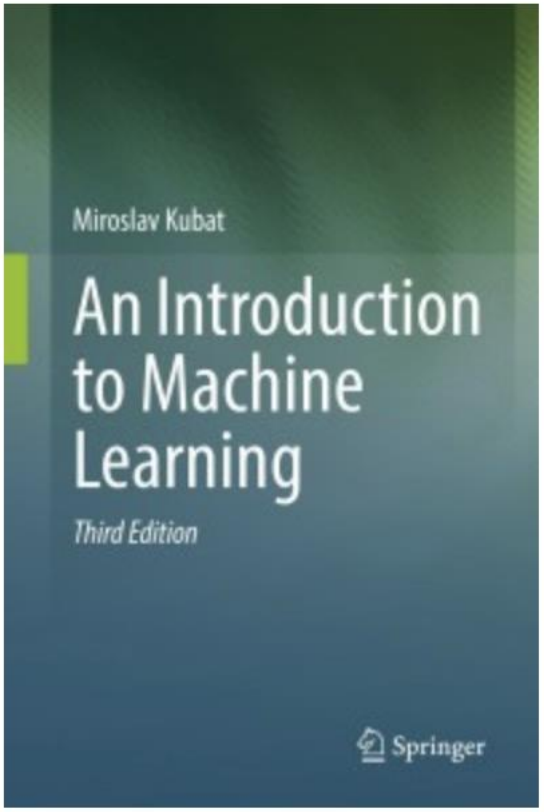
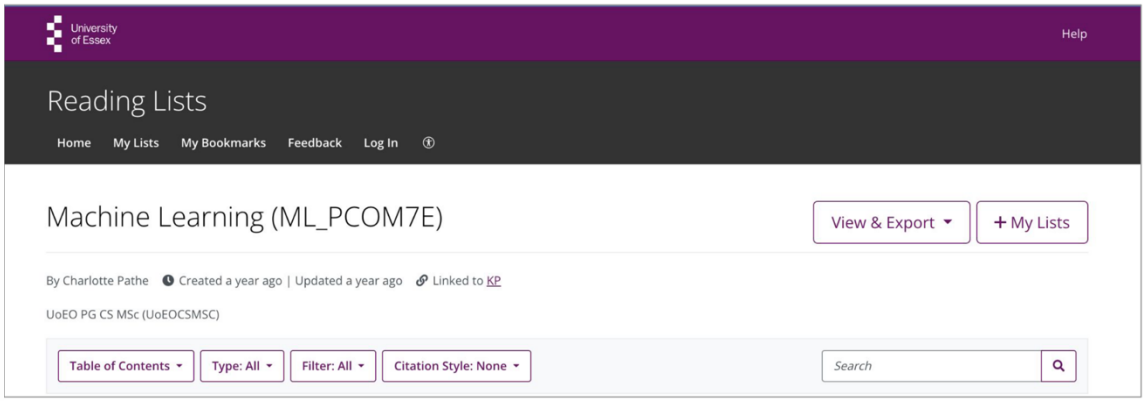
Our Textbooks & Resources



Access My eBook

https://rl.talis.com/3/essex/lists/8B446180-9F72-E5CB-BB1E-34A56D0DF456.html?lang=en

This links directly to your reading list



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Unit 8 Readings

Unit 8

Required Reading

An introduction to machine learning
Book - by Miroslav Kubát - 2017 - Second edition - **Essential**
4.5
VIEW ONLINE

10 Business Applications of Neural Network
Webpage - by Mach, P - 2021 - **Essential**
VIEW ONLINE

Snapshot Paper - AI and Personal Insurance - GOV.UK
Webpage - **Essential**
VIEW ONLINE

Robo-writers: the rise and risks of language-generating AI
Webpage - by Hutson, M. - **Essential**
VIEW ONLINE

Can we open the black box of AI? in Nature
Article - by Davide Castelvecchi - 10/2016 - **Essential**
VIEW ONLINE

Explained: Updating Weights with Gradient Descent & Backpropagation
Webpage - by Mayo, M - 2017 - **Essential**
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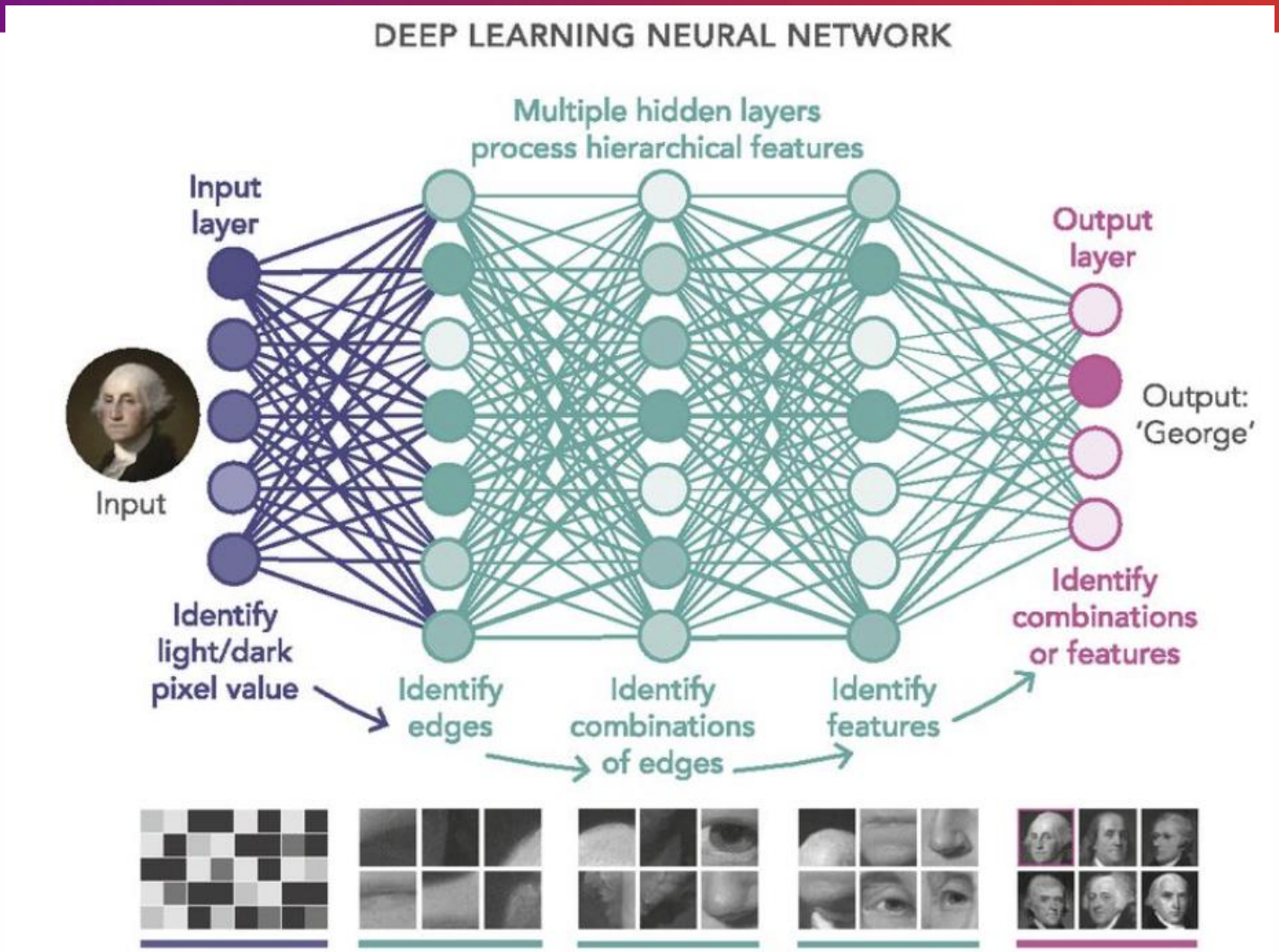
An Artificial Neural Network (ANN) is a computational model inspired by the way biological neural networks in the human brain process information. ANNs consist of interconnected groups of artificial neurons, which work in unison to solve specific problems.

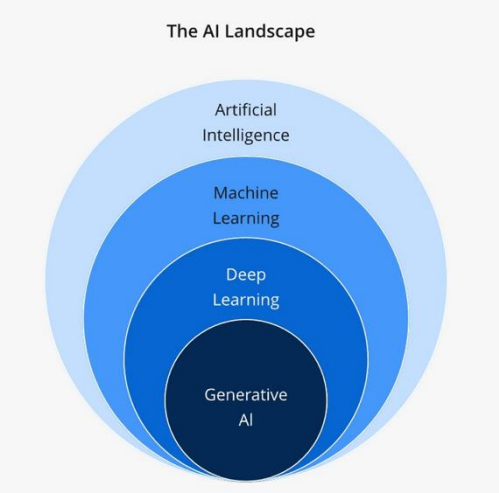


Machine Learning vs Deep Learning

If you have modest resources → stick with ML

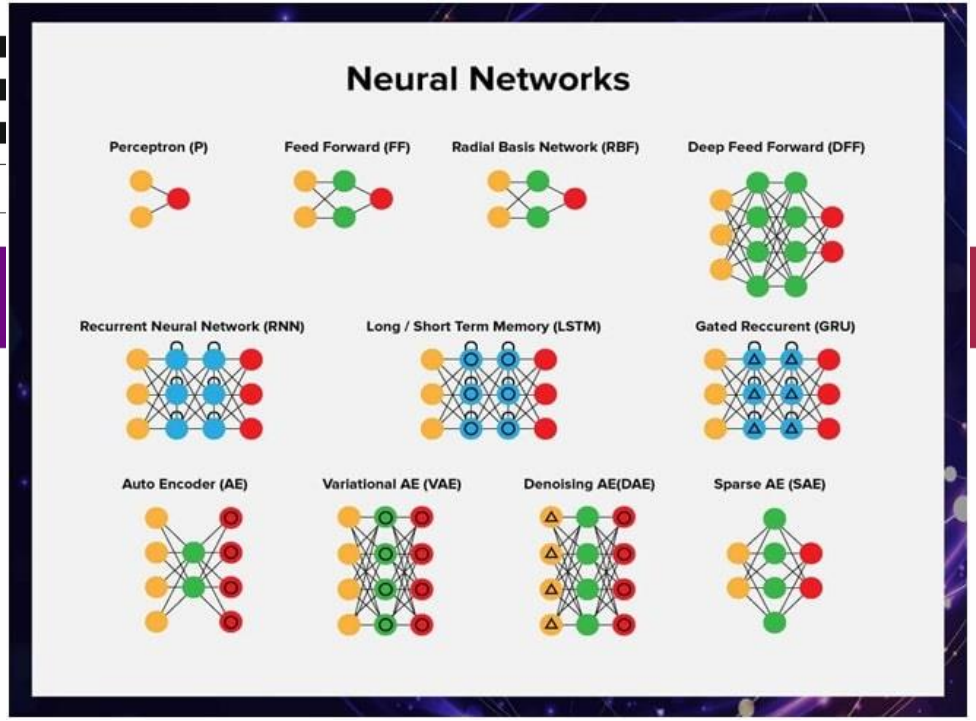
If you have access to GPU/TPU → opt for DL

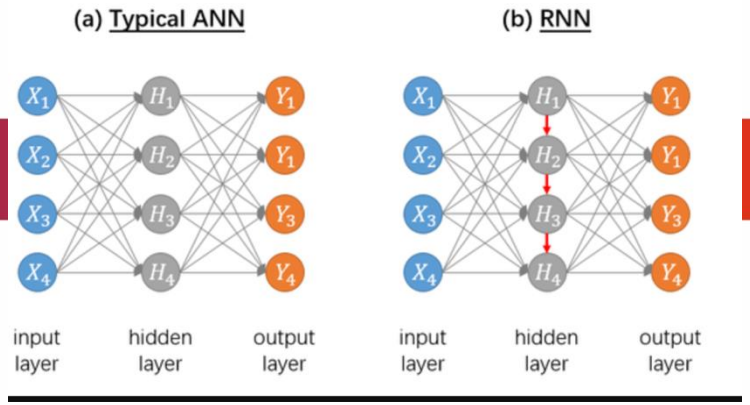
Machine Learning	Deep Learning
Optimal data volumes	
Thousands of data points	Big data: millions of data points
Outputs	
Numerical value, like classification or score	Anything from numerical values to free-form element, such as free text and sounds
How it works	
Uses various types of automated algorithms that learn to model functions and predict future actions from data	Uses neural networks that pass through many processing layers to interpret data features and relations
How it's managed	
Algorithms are detected by data analysts to examine specific variable in data sets	Algorithms are largely self-directed on data analysis once they're put into production



If transparency is crucial → ML is the way to go

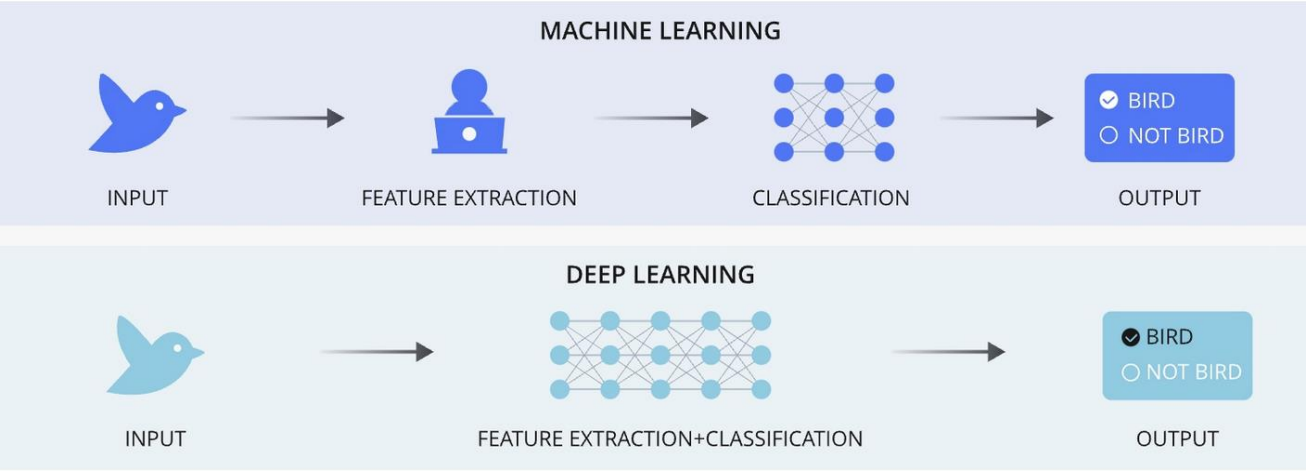
If accuracy is your top priority → DL is better





DL is basically ML with neural networks as the model.

So the same concepts apply: some problems are classification, some are regression.



https://www.fs.com/blog/artificial-intelligence-vs-machine-learning-vs-deep-learning-vs-generative-ai-key-differences-explained-24134.html

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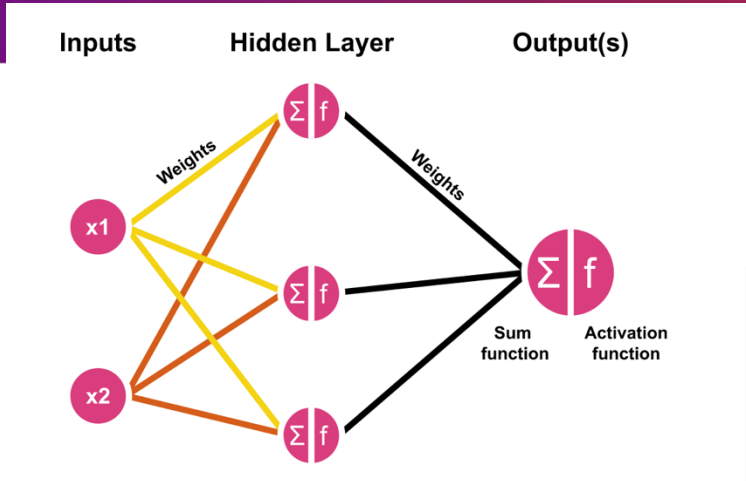
Multi-Layer Perceptron (Multi-Layer Neural Networks)



Activation functions	Weight update (XOR operator)	Error functions
Gradient descent	Backpropagation	Implementation of a multi-layer perceptron from scratch using Python and Numpy

1. **simple_perceptron.ipynb**
2. **perceptron_AND_operator.ipynb**
3. **multi-layer Perceptron.ipynb** (the Sigmoid Function)

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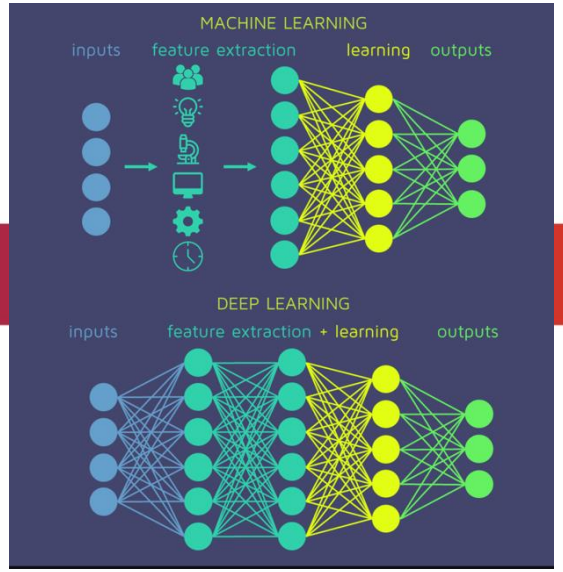
$$z = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$



Components

1.Neurons (Nodes): Basic processing units of the network, each performing a simple computation.

2.Layers:

1. **Input Layer**: Receives the input data.
2. **Hidden Layers**: Intermediate layers where the *actual processing is done* through weighted connections.
3. **Output Layer**: Produces the final output of the network.

3.Weights/ biases: Parameters that adjust the input's influence on the neuron's output.

4.Activation Functions: Functions that introduce non-linearity into the network, helping it learn complex patterns. Common activation functions include ReLU, Sigmoid, and Tanh.

Training Process

1.Initialization: Randomly initializing weights.

2.Forward Propagation: Passing input data through the network to get the output.

3.Loss (Error) Calculation: Computing the error by comparing the predicted output with the actual target values using a loss function.

4.Backpropagation: Algorithm used to compute the gradient of the loss function with respect to each weight in the network. It adjusts the weights based on the error to reduce it.

5.Iteration: Repeating the process (forward propagation, error calculation, and backpropagation) for many iterations or epochs until the error is minimized.

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Backpropagation in Artificial Neural Networks (ANNs)

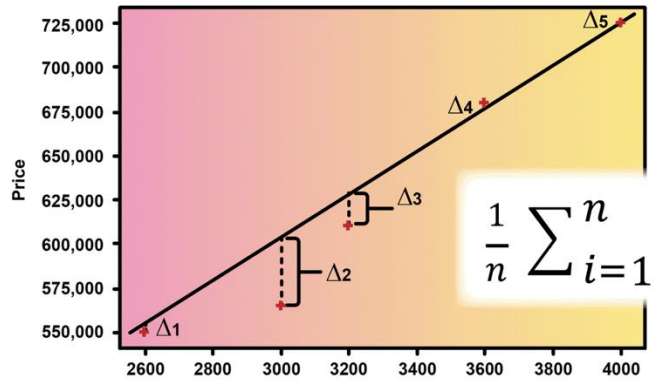
- A mechanism used to update the weights using gradient descent.
- It calculates the gradient of the error function with respect to the neural network's weights.

In order for a **neural networks** to **learn**, **weights associated** with **neuron connections** must be **updated after forward passes of data through the network.**

The error can be defined as the difference between the actual output and the predicted output at each data point. Mean squared error or MSE defines this value with the equation. Since the value of the regression straight line can be represented by the straight line formula y = mx + b, the equation for MSE can be rewritten as shown.

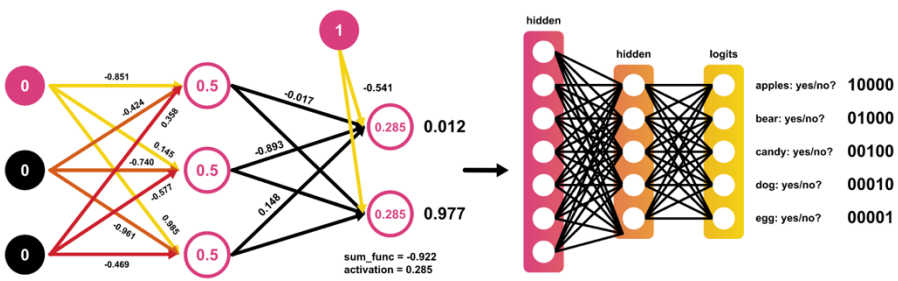
Since we are talking about the **difference** between **actual** and **predicted values**, the error would be a **useful measure** here.

Each neuron will **require** that their **respective error** be **sent backward through the network to them**, in order to **facilitate the update process**; hence the name, **backpropagation of error**.



$$\frac{1}{n}\sum_{i=1}^{n}(\Delta i)^2$$

So far, we have covered a NN with one hidden layer. In reality, NNs have multiple hidden layers through which backpropagation occurs.



Loss calculation:

>> Compare prediction to the ground truth (label) using your loss function.

Example: cat image predicted as 80% dog, 20% cat → high loss.

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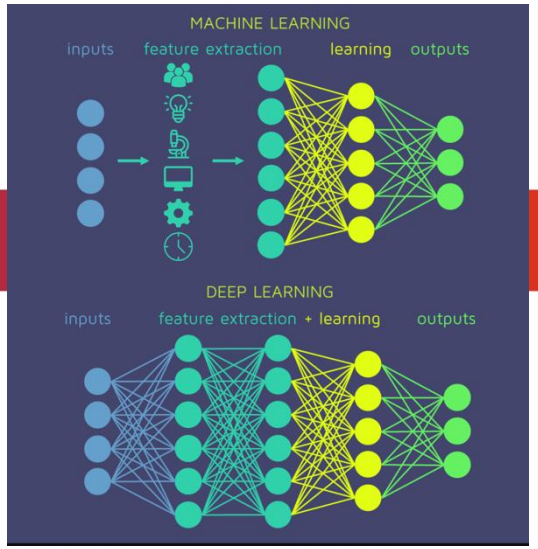
*in DL, the architecture is mostly the same (convolutions, hidden layers), but **the last layer + activation + loss** are what turn the network into a classifier or regressor.*



What you can choose/tweak:

Architecture: algorithms, number of layers, type of layers, kernel size in convolutions, etc.

Activation function: decides how signals flow (**ReLU**, Sigmoid, Tanh, etc.).

Loss function: defines what “error” means (cross-entropy for classification, MSE for regression).

Optimiser: how weights are updated (SGD, Adam, RMSprop).

Learning rate (LR): size/type of steps when adjusting weights (cosine decay, step decay etc.).

Batch size: number of samples processed before weight update.

Number of epochs: how many times the model sees the whole dataset.

What you cannot directly set:

Weights: initial weights are usually random (with smart initialization etc.). They change automatically during training.

Biases: often start at zeroes, then shift the activation function.

Features learned: you don’t tell the CNN to look for "cat ears" — it discovers features itself through training.

!! If dataset is small: keep a bigger validation set
!! Also consider data augmentation

**** Training set (largest part)**

Used to update weights (via forward pass, loss, backpropagation).

Usually 60–80% of your total data.

**** Validation set**

Not used to update weights, only used to tune hyperparameters (learning rate, architecture, regularization).

Prevents overfitting by checking how the model performs on unseen data during training.

Usually 10–20% of your data.

**** Test set**

Used only once after training/validation are done, to estimate final performance.

Simulates “real-world” unseen data.

Usually 10–20% of your data.

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Unit 8 Seminar

Title: Emerging Research in ANN

Read the article by Mach (2021) and pick one area that interests you or have experience with. Note down your views on the use and impact of ANN on the selected area.

Read the snapshot paper by the Centre for Data Ethics and Innovation on AI and Personal Insurance. As a customer, note down your concerns about use of AI in personal insurance.

Discuss your views and concerns as a team. Be prepared to share your team's thoughts on both issues with the rest of the class during the seminar.

There will also be an opportunity to discuss your progress on the team project during the seminar.

Learning Outcomes

- Apply and critically appraise machine learning techniques to real-world problems, particularly where technical risk and uncertainty is involved.
- Systematically develop and implement the skills required to be effective member of a development team in a virtual professional environment, adopting real-life perspectives on team roles and organisation.

https://www.gov.uk/government/publications/cdei-publishes-its-first-series-of-three-snapshot-papers-ethical-issues-in-ai/snapshot-paper-ai-and-personal-insurance

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Unit 8 Seminar

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https://www.gov.uk/government/publications/cdei-publishes-its-first-series-of-three-snapshot-papers-ethical-issues-in-ai/snapshot-paper-ai-and-personal-insurance

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Independent report

Snapshot Paper - AI and Personal Insurance

Published 12 September 2019

Artificial intelligence (AI) is expected to alter several dimensions of the personal insurance industry, including customer onboarding (e.g. by powering customer service chatbots), pricing (e.g. by enabling more precise risk assessments) and claims management (e.g. by screening out fraudulent claims).

Looking further ahead, AI could one day enable insurers to offer novel forms of advisory services that help customers to live healthier and safer lives, for example by recommending safer driving routes or by flagging early signs of damage in the home.

The industry is still in the nascent stages of AI adoption. Incumbents have found it challenging to marry this technology with their legacy infrastructure, as well as to find the talent required to take forward innovation programmes. Yet insurance leaders are confident that AI will soon be embedded across their value chains.

Critics say that AI could lead to detrimental outcomes for customers, particularly where it allows or requires:

1. the **collection and sharing of large data troves**, which could impinge on privacy if done without the express consent of customers
2. **hyper personalised risk assessments**, which could leave some individuals 'uninsurable' by revealing previously unseen indicators of risk
3. **new forms of nudging**, where insurers use AI to alter the behaviour of customers in a way that could be viewed as intrusive

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Collaborative Discussion 2: Legal and Ethical views on ANN applications

Forum Subscriptions Reports Export ···

Mark as done

This discussion will last for 3 weeks covering units 8, 9 and 10. Ensure you include appropriate citations and references in all your posts (this is not included in the word count for each week's post/submission).

Discussion Topic

Read/listen to the article by Hutson (2021) in Nature on Robo-writers. This week, post your thoughts on the risks and benefits of the use AI writers at different levels, from administrative work to creative writing.

You should demonstrate that you understand the topic covered and ensure you use references to academic literature (journals, books, reports, etc.).

Learning Outcomes

- Apply and critically appraise machine learning techniques to real-world problems, particularly where technical risk and uncertainty is involved.

Assignment Guidance

- Your initial posting should respond to the question and be at least 200 words long. This should be labelled as '**Initial Post**'
- You will then respond to **at least 2 of your peers' posts** in unit 9 (each labelled as '**Peer Response**'). To guide your responses, look at the guidelines for the peer review process on the **Department's homepage**. Focus on the possible measures that could have been put in place in order to prevent the incidents highlighted by your peers. Please try to limit your response posts to **200-300 words maximum**, so that others may be encouraged to reflect on, and respond to your ideas.
- In Unit 10, you should provide a summary post based on your initial post, the feedback from your peers and the content of the three units. Please label this as '**Summary Post**'. It should be **300 words**.
- Referencing: When you have referred to other authors thoughts, ideas and opinions in your posts, you must reference the author as you would in an academic assignment using the UoEO Harvard reference style.
- You will be not be assessed on your contribution to this forum throughout the module, but the tutor will post group feedback via a module announcement.
- This activity forms a component of your e-portfolio which you will submit in unit 12. All e-portfolio activities are intended to demonstrate your ability and strengths through evidence and reflection.

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Next Session

Assignment 2 Topic: Neural Network Models for Object Recognition

CIFAR-10

Please access the dataset here

CIFAR-10 is a cornerstone dataset for machine learning research, published in 2009, it provided a standard for algorithms in image recognition.

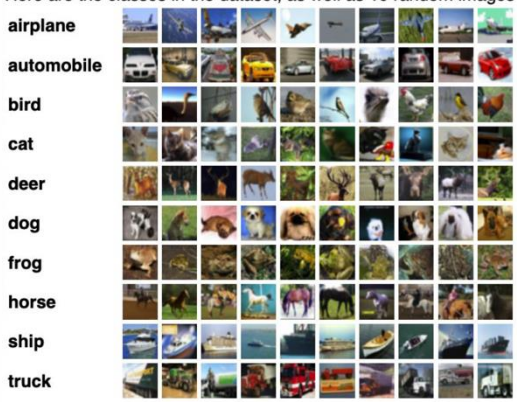
Its complexity is intermediate, making it more challenging than MNIST but less daunting than ImageNet, thus serving as a vital stepping stone in the progression of computer vision studies.

The CIFAR-10 dataset

The CIFAR-10 dataset consists of 60000 32x32 colour images in 10 classes, with 6000 images per class. There are 50000 training images ar 10000 test images.

The dataset is divided into five training batches and one test batch, each with 10000 images. The test batch contains exactly 1000 randomly-selected images from each class. The training batches contain the remaining images in random order, but some training batches may contain more images from one class than another. Between them, the training batches contain exactly 5000 images from each class.

Here are the classes in the dataset, as well as 10 random images from each:



The classes are completely mutually exclusive. There is no overlap between automobiles and trucks. "Automobile" includes sedans, SUVs, things of that sort. "Truck" includes only big trucks. Neither includes pickup trucks.

Download

If you're going to use this dataset, please cite the tech report at the bottom of this page.

Version	Size	md5sum
CIFAR-10 python version	163 MB	c58f30108f718f92721af3b95e74349a
CIFAR-10 Matlab version	175 MB	70270af85842c9e89bb428ec9976c926
CIFAR-10 binary version (suitable for C programs)	162 MB	c32a1d4ab5d03f1284b67883e8d87530

Questions?